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Assignment No:1 Demonstrate fundamental concepts of Java

1. Write a program to calculate perimeter and area of rectangle. (hint : area = length * breadth , perimeter=2*(length+breadth))

Assignment_2 > RectangleCalculation.java > ...

```java
import java.util.Scanner;
public class RectangleCalculation {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print(s: "Enter length of rectangle: ");
        double length = sc.nextDouble();

        System.out.print(s: "Enter breadth of rectangle: ");
        double breadth = sc.nextDouble();

        double area = length * breadth;
        double perimeter = 2 * (length + breadth);

        System.out.println("Area of Rectangle = " + area);
        System.out.println("Perimeter of Rectangle = " + perimeter);

        sc.close();
    }
}
```

Output:-

```
PS G:\My Drive\MCA Documents\SEM 2\Java\assignment_2> java RectangleCalculation.java
Enter length of rectangle: 23
Enter breadth of rectangle: 54
Area of Rectangle = 1242.0
Perimeter of Rectangle = 154.0
```

2. Write a menu driven program to perform the following operations
   i. Calculate the volume of cylinder. (hint : Volume: $\pi \times r^2 \times h$)
   ii. Find the factorial of given number.
   iii. Check the number is Armstrong or not.
   iv. Exit

Assignment_2 > MenuDrivenProgram.java > MenuDrivenProgram > main(String[])

```java
import java.util.Scanner;
public class MenuDrivenProgram {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int choice;
        do {
            System.out.println(x: "\n===== MENU =====");
            System.out.println(x: "1. C1alculate Volume of Cylinder");
            System.out.println(x: "2. Find Factorial");
            System.out.println(x: "3. Check Armstrong Number");
            System.out.println(x: "4. Exit");
            System.out.print(s: "Enter your choice: ");
            choice = sc.nextInt();
            switch (choice) {
                case 1:
                    // Volume of Cylinder
                    System.out.print(s: "Enter radius: ");
                    double radius = sc.nextDouble();
                    System.out.print(s: "Enter height: ");
                    double height = sc.nextDouble();
                    double volume = Math.PI * radius * radius * height;
                    System.out.println("Volume of Cylinder = " + volume);
                    break;
                case 2:
                    // Factorial
                    System.out.print(s: "Enter a number: ");
                    int num = sc.nextInt();
                    long fact = 1;
```

```java
                long fact = 1;
                for (int i = 1; i <= num; i++) {
                    fact *= i;
                }
                System.out.println("Factorial of " + num + " = " + fact);
                break;
            case 3:
                // Armstrong Number Check
                System.out.print(s: "Enter a number: ");
                int number = sc.nextInt();
                int original = number;
                int sum = 0, remainder;

                while (number != 0) {
                    remainder = number % 10;
                    sum += remainder * remainder * remainder;
                    number /= 10;
                }
                if (sum == original)
                    System.out.println(original + " is an Armstrong Number.");
                else
                    System.out.println(original + " is NOT an Armstrong Number.");
                break;
            case 4:
                System.out.println(x: "Exiting Program...");
                break;
            default:
```

```java
            default:
                System.out.println(x: "Invalid choice! Please try again.");
        }
    } while (choice != 4);
    sc.close();
}
}

```

Output:-

```
PS G:\My Drive\MCA Documents\SEM 2\Java\assignment_2> java MenuDrivenProgram.java

 ===== MENU =====
 1. C1alculate Volume of Cylinder
 2. Find Factorial
 3. Check Armstrong Number
 4. Exit
 Enter your choice: 3
 Enter a number: 153
 153 is an Armstrong Number.
```

3. Write a program to accept the array element and display in reverse order.

Assignment_2 > ReverseArray.java > ReverseArray

```java
import java.util.Scanner;
class ReverseArray {
    Run | Debug
    public static void main(String[] args) {
        int arr[] = new int[5];
        Scanner sc = new Scanner(System.in);
        System.out.println(x: "Enter 5 elements:");
        for (int i = 0; i < 5; i++) {
            arr[i] = sc.nextInt();
        }
        System.out.println(x: "Array elements in reverse order:");
        for (int i = 4; i >= 0; i--) {
            System.out.println(arr[i]);
        }
    }
}
```

Output:-

```
PS G:\My Drive\MCA Documents\SEM 2\Java\assignment_2> java ReverseArray.java
Enter 5 elements:
15
20
44
10
12
Array elements in reverse order:
12
10
44
20
15
```

4. Write a menu driven program to perform the following operations on multidimensional array ie matrix :
   i. Addition
   ii. Multiplication
   iii. Transpose of any matrix.
   iv. Exit

Assignment_2 > MatrixOperations.java > ...

```java
import java.util.Scanner;

class MatrixOperations {
    Run | Debug
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        int choice;
        do {
            System.out.println(x: "\nMenu");
            System.out.println(x: "1. Addition");
            System.out.println(x: "2. Multiplication");
            System.out.println(x: "3. Transpose");
            System.out.println(x: "4. Exit");
            System.out.print(s: "Enter your choice: ");
            choice = sc.nextInt();
            switch (choice) {
                case 1: // Matrix Addition
                    int a[][] = new int[2][2];
                    int b[][] = new int[2][2];
                    int c[][] = new int[2][2];
                    System.out.println(x: "Enter first matrix:");
                    for (int i = 0; i < 2; i++)
                        for (int j = 0; j < 2; j++)
                            a[i][j] = sc.nextInt();
                    System.out.println(x: "Enter second matrix:");
                    for (int i = 0; i < 2; i++)
                        for (int j = 0; j < 2; j++)
                            b[i][j] = sc.nextInt();
                    for (int i = 0; i < 2; i++)
                        for (int j = 0; j < 2; j++)
                            c[i][j] = a[i][j] + b[i][j];
                    System.out.println(x: "Addition of matrix:");
                    for (int i = 0; i < 2; i++) {
                        for (int j = 0; j < 2; j++)
                            System.out.print(c[i][j] + " ");
                        System.out.println();
                    }
                    break;
                case 2: // Matrix Multiplication
                    int m1[][] = new int[2][2];
                    int m2[][] = new int[2][2];
                    int m3[][] = new int[2][2];
                    System.out.println(x: "Enter first matrix:");
                    for (int i = 0; i < 2; i++)
                        for (int j = 0; j < 2; j++)
                            m1[i][j] = sc.nextInt();
                    System.out.println(x: "Enter second matrix:");
```

```java
                System.out.println(x: "Enter second matrix:");
                for (int i = 0; i < 2; i++)
                    for (int j = 0; j < 2; j++)
                        m2[i][j] = sc.nextInt();
                for (int i = 0; i < 2; i++)
                    for (int j = 0; j < 2; j++) {
                        m3[i][j] = 0;
                        for (int k = 0; k < 2; k++)
                            m3[i][j] += m1[i][k] * m2[k][j];
                    }
                System.out.println(x: "Multiplication of matrix:");
                for (int i = 0; i < 2; i++) {
                    for (int j = 0; j < 2; j++)
                        System.out.print(m3[i][j] + " ");
                    System.out.println();
                }
                break;
            case 3: // Transpose
                int t[][] = new int[2][2];
                System.out.println(x: "Enter matrix:");
                for (int i = 0; i < 2; i++)
                    for (int j = 0; j < 2; j++)
                        t[i][j] = sc.nextInt();
                System.out.println(x: "Transpose of matrix:");
                for (int i = 0; i < 2; i++) {
                    for (int j = 0; j < 2; j++)
                        System.out.print(t[j][i] + " ");
                    System.out.println();
                }
                break;
            case 4:
                System.out.println(x: "Exiting program...");
                break;
            default:
                System.out.println(x: "Invalid choice");
        }
    } while (choice != 4);
    sc.close();
  }
}
```

Output:-

```
PS G:\My Drive\MCA Documents\SEM 2\Java\assignment_2> java MatrixOperations.java

 Menu
 1. Addition
 2. Multiplication
 3. Transpose
 4. Exit
 Enter your choice: 1
 Enter first matrix:
 12
 14
 12
 24
 Enter second matrix:
 36
 45
 12
 42
 Addition of matrix:
 48 59
 24 66
```

5. Write a program to accept n names of country and display them in descending order.

Assignment_2 > CountryNames.java > CountryNames

```java
import java.util.Scanner;

class CountryNames {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print(s: "Enter number of countries: ");
        int n = sc.nextInt();
        sc.nextLine();
        String country[] = new String[n];
        System.out.println(x: "Enter country names:");
        for (int i = 0; i < n; i++) {
            country[i] = sc.nextLine();
        }
        for (int i = 0; i < n; i++) {
            for (int j = i + 1; j < n; j++) {
                if (country[i].compareTo(country[j]) < 0) {
                    String temp = country[i];
                    country[i] = country[j];
                    country[j] = temp;
                }
            }
        }
        System.out.println(x: "Countries in descending order:");
        for (int i = 0; i < n; i++) {
            System.out.println(country[i]);
        }
        sc.close();
    }
}
```

Output:-

```
PS G:\My Drive\MCA Documents\SEM 2\Java\assignment_2> java CountryNames.java
Enter number of countries: 5
Enter country names:
India
Russia
South Africa
Brazil
China
Countries in descending order:
South Africa
Russia
India
China
Brazil
```

6. Write a menu driven program to perform the following operations on 2D array:
   i. Sum of diagonal elements
   ii. Sum of upper diagonal elements
   iii. Sum of lower diagonal elements
   iv. Exit

Assignment_2 > MatrixMenu.java > MatrixMenu

```java
import java.util.Scanner;

class MatrixMenu {
    Run | Debug
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        int n, choice;
        System.out.print(s: "Enter size of matrix: ");
        n = sc.nextInt();
        int a[][] = new int[n][n];
        System.out.println(x: "Enter matrix elements:");
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                a[i][j] = sc.nextInt();
            }
        }
        do {
            System.out.println(x: "\nMenu");
            System.out.println(x: "1. Sum of Diagonal Elements");
            System.out.println(x: "2. Sum of Upper Diagonal Elements");
            System.out.println(x: "3. Sum of Lower Diagonal Elements");
            System.out.println(x: "4. Exit");
            System.out.print(s: "Enter your choice: ");
            choice = sc.nextInt();
            switch (choice) {
                case 1:
                    int diagSum = 0;
                    for (int i = 0; i < n; i++) {
                        diagSum += a[i][i];
                    }
                    System.out.println("Sum of diagonal elements = " + diagSum);
                    break;
                case 2:
                    int upperSum = 0;
                    for (int i = 0; i < n; i++) {
                        for (int j = 0; j < n; j++) {
                            if (i < j) {
                                upperSum += a[i][j];
                            }
                        }
                    }
```

```java
                System.out.println("Sum of upper diagonal elements = " + upperSum);
                break;
            case 3:
                int lowerSum = 0;
                for (int i = 0; i < n; i++) {
                    for (int j = 0; j < n; j++) {
                        if (i > j) {
                            lowerSum += a[i][j];
                        }
                    }
                }
                System.out.println("Sum of lower diagonal elements = " + lowerSum);
                break;

            case 4:
                System.out.println(x: "Exiting program...");
                break;

            default:
                System.out.println(x: "Invalid choice");
        }
    } while (choice != 4);
    sc.close();
}
}
```

Output:-

```
PS G:\My Drive\MCA Documents\SEM 2\Java\Assignment_2> java MatrixMenu.java
Enter size of matrix: 2
Enter matrix elements:
1
2
3
4

Menu
1. Sum of Diagonal Elements
2. Sum of Upper Diagonal Elements
3. Sum of Lower Diagonal Elements
4. Exit
Enter your choice: 1
Sum of diagonal elements = 5
```